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The apparent polar wander path of the European plate in Upper Triassic–Lower Jurassic times and the Liassic intraplate fracturing of Pangaea: new palaeomagnetic constraints from NW France and SW Germany

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SUMMARY

In order to constrain the European plate motions during Triassic–Lower Jurassic times, new palaeomagnetic data were acquired from various Triassic–Liassic sediments of northeastern France and southwestern Germany. A Carnian palaeopole 49°N/131°E, $K=193$, $A_{95}=7°$, $N=4$ was obtained for Keuper marls with consistent directions of normal and reversed polarities. Rhaetian sandstones, also with both polarities, and Carnian dolomites with a magnetization of probable secondary origin, provide a mean pole 51°N/112°E, $K=109$, $A_{95}=6°$, $N=6$, more westerly than that given above. The Lower Hettangian, sampled in limestones of the Xeuilley quarry near Nancy shows a N–R–N polarity sequence with well-defined anti-parallel directions. The mean Hettangian pole is 55°N/100°E, $K=123$, $A_{95}=11°$, $N=3$. Induan sandstones from the northern Vosges show two distinct overprints, one consistent with the Ladinian–Carnian magnetizations already observed in the same region, the second consistent with the recalculated mean Lower Jurassic pole of Europe. Five other sites with different rock types also exhibit late overprints with Lower Jurassic directions. The mean pole calculated from these overprints is 74°N/124°E, $K=125$, $A_{95}=6°$, $N=6$.

The revised apparent polar wander path (APWP) presented here shows that for the mid-Permian to the earliest Jurassic the palaeopoles of Europe stayed around 50° latitude, while the longitudes drifted from 185°E to 100°E. The previously established Upper Triassic–Lower Jurassic loop of the APWP is shifted westwards by about 25°. The revised APWP involves the following plate motions. After the late Alleghanian tectonic phase, which marked the end of the Variscan orogeny, central western Europe rotated counterclockwise during about 70 Myr, up to the earliest Jurassic. The south–north drift in latitude by 35° (nearly 4000 km) remained regular from 320 to 205 Ma, with a constant rate of 3.4 cm yr^{-1}. Around 205 Ma, this motion ceased, probably because of a collision somewhere in the north of Eurasia. Then, in the Lower Liassic, the plate underwent a rapid counterclockwise rotation with a Eulerian pole located in northeastern Asia, synchronous with fracturing in the Atlantic and the Neotethys areas. It is proposed that the intraplate fracturing of Pangaea was initiated by the rapid change of the megacontinent motion at the end of the Trias.

Key words: France, Germany, Jurassic, palaeomagnetism, Trias.

INTRODUCTION

This paper reports palaeomagnetic data that were collected in order to provide a precise Triassic–Early Jurassic apparent polar wander path (APWP) of the European plate and to fill the gaps in the available palaeomagnetic database in order to trace the evolution of the plate in detail from the latest stages of the Variscan orogeny up to the Jurassic opening of the Atlantic. It extends previous works on the Permian–Early Triassic APWP, which (1) demonstrated a counterclockwise hairpin in the European APWP in the mid-Permian, resulting from a drastic change in the European plate motion (Edel &

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Schneider 1995) and (2) provided a palaeopole for the Lower Anisian (Edel 1993).

Following the large clockwise rotation of the European plate during the Late Variscan convergence from the mid-Carboniferous to the mid-Permian, palaeomagnetic data suggest a counterclockwise rotation in the Upper Permian and most of the Trias (Westphal *et al.* 1986; Van der Voo 1990; Théveniaut 1993; Beck 1994). The main objective of this study is to document the next significant event in the European APWP, which corresponds to a clockwise loop of the curve in Upper Triassic–Lower Jurassic times. This critical period, which marks the beginning of rifting in the central part of the future Atlantic, and of the formation of the Tethys, is unfortunately ill-constrained by palaeomagnetic data. For the Middle and Upper Triassic, as well as for the Lower Jurassic, data are scarce and quality factors are mostly moderate (Van der Voo 1990). The data recently obtained from dolerites in Brittany raise some problems because they are at variance with the previous results (Sichler & Perrin 1993). Therefore, extensive sampling was carried out in Upper Triassic and Lower Jurassic series from the eastern part of the Paris Basin and from the south German basin, with emphasis on the Keuper series. A large part of the rocks collected does not contain stable magnetizations, or they are below the noise level of our magnetometer. Nevertheless, about half of the stratigraphic levels sampled provided well-defined magnetizations that can be used to define the European APWP for the Early Mesozoic better.

GEOLOGICAL SETTING AND SAMPLING

Triassic strata of eastern France and Germany (the German Triassic) consist of a basal detrital unit (Buntsandstein) overlain by a carbonaceous unit (Muschelkalk) and a marly dolomitic, marly gypsiferous and marly sandy unit (Keuper). The Buntsandstein sequence contains coarse-grained sandstones interrupted by less-abundant conglomeratic levels. The depositional environment of the Buntsandstein is predominantly alluvial with some minor aeolian episodes. In the upper Buntsandstein unit, the alluvial facies are progressively replaced by deltaic, estuary and near-shore facies (Grès à Voltzia; NVS, Table 1) (Edel 1993). Coarse sandstones grade into fine sandstones and pelites. These rocks mark the transition to the marine environment characteristic of the Muschelkalk. In the field, the boundary between the fluvio-estuarine and fluvio-deltaic and the marine units is characterized by a basal transgression (onlap) surface.

The Muschelkalk sequence is characterized by three depositional episodes: two marine episodes with abundant echinoderm remains (Lower and Upper Muschelkalk) are separated by a confined to evaporitic episode. A palaeomagnetic pole was obtained from the upper Muschelkalk in the Heming quarry by Théveniaut *et al.* (1992) (Fig. 1).

The Upper Triassic Keuper facies mark the return to a transitional to continental depositional environment consisting mostly of evaporites and continental to marine near-shore sandstones that occasionally interrupt the monotony of marly dolomitic and marly gypsiferous deposits (reed sandstones of the Middle Keuper and Rhaetian sandstones). The Lias marks the return to a marine episode and the beginning of the major Mesozoic transgression.

The Germanic Trias units do not have an exact geochronological correspondence, and on a large scale their facies are often diachronous from east to west. The stratigraphic subdivision of the German Trias is based on a very detailed definition of the lithostratigraphic units. The correspondence between the geological lithostratigraphic scale of the German basin (Hagdorn & Simon 1985) and the geological timescale

Table 1. General correlation of the German facies with the Triassic stratigraphic position of sites, after Odin (1994) and Hagdorn & Simon (1985).

INTERNATIONAL GEOLOGICAL TIME SCALE				GERMAN BASIN STRATIGRAPHY		SAMPLING SITES
JURASSIC	LIAS	175	TOARCIAN			
		184	PLIENSBACHIAN			
		191	SINEMURIAN			CHS 2
		200	HETTANGIAN			CHS 2
		203				PFD 2 - XEU1 - XEU2
TRIASSIC	LATE		RHAETIAN	KEUPER	*Rät*	SNP - STG - PFD 1
			NORIAN		*Knollenmergel*	
					Stubensandstein	BEG
					Kieselsandstein	
		220			*Rote Wand*	
			CARNIAN		*Dunkle Mergel*	CHM
					Schilfsandstein	CHS 1-ETD
					Gipskeuper	OSH 1- OSH 2 - ENT 1 - ENT 2
		230	LADINIAN		*Lettenkeuper*	ESA
	MID	233		MUSCHELKALK	*Upper*	HEM
			ANISIAN		*Middle*	
					Lower	
	EARLY	240	OLENEKIAN	BUNTSANDSTEIN	*Upper (Röt)*	NVS
		Ma			*Middle*	
			INDUSIAN		*Lower*	NST
		250				

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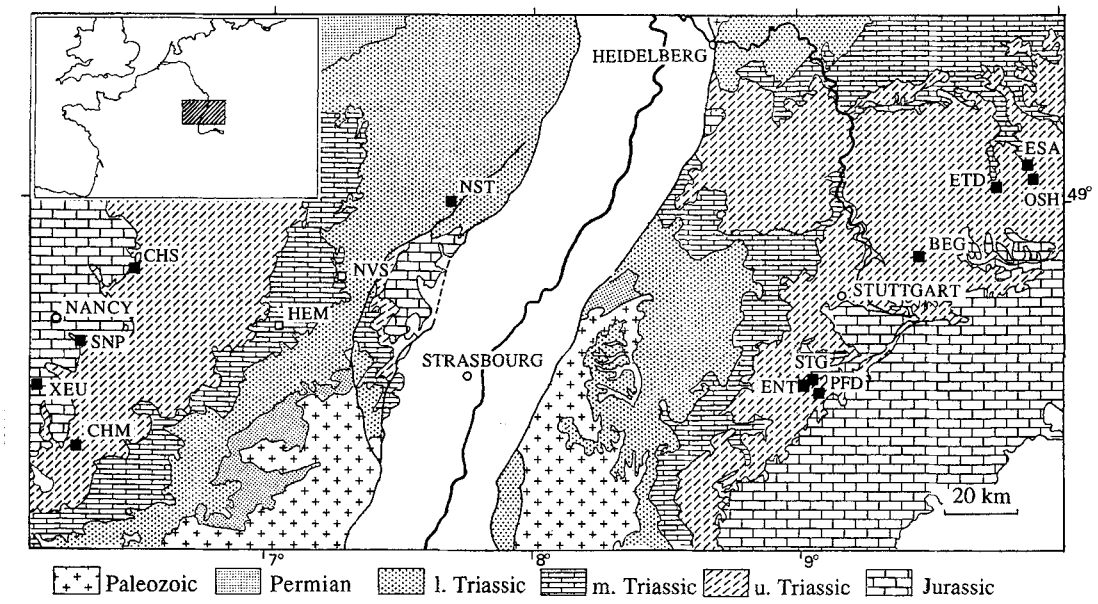


Figure 1. Geological sketch map of NE France and SW Germany showing the locations of palaeomagnetic sites. Black squares indicate localities of this study. HEM shows the Heming Muschelkalk quarry (Théveniaut *et al.* 1992) and NVS the Anisian sandstones of the northern Vosges (Edel 1993).

(Odin 1994) is given in Table 1. The German nomenclature is based on rock colour (Rote Wand = red wall; Dunkle Mergel = dark marl), fossil contents (Schilfsandstein = reed sandstone) and petrographic peculiarities (Gipskeuper = gypsum-rich unit of the Keuper; Knollen Mergel = chalk marl).

Sampling was performed in most units representative of the Triassic Germanic facies and in the Lower Lias. Interpretable results with primary or secondary magnetizations were obtained in about half of the localities investigated. The samples were cored and/or collected as hand samples in quarries (from west to east: XEU, SNP, NST, ENT, STG, PFD, BEG, ETD, ESA, OSH) and recent road cuts (CHS, CHM) (Fig. 1; Tables 1 and 2).

RESULTS

Palaeomagnetic and rock-magnetic experiments

The results presented here concern sites that yielded well-defined characteristic magnetizations of geological significance, primary or secondary. A quarry or a road cut is considered as a locality and a site represents a stratigraphic unit. 8–25 samples were taken as hand samples or cored at a site and each core was cut into 2 or 3 specimens. The direction and intensity of natural remanent magnetization were measured on a modified Digico spinner magnetometer with a maximum noise level at 0.03 mA m^{-1}. Low-field susceptibility and anisotropy of magnetic susceptibility measurements were also performed on Digico devices. Isothermal remanent magnetization (IRM) acquisition and thermal demagnetization of three-component IRM (Lowrie 1990), and analysis of polished thin sections were performed in order to identify the magnetic phases. Specimens were thermally demagnetized using home-made furnaces. Characteristic directions of remanence were computed using line fitting to segments in Zijderveld diagrams with at least three points per line (Kent, Briden & Mardia 1983; Kirschvink 1980).

Because of magnetizations that were too low or because the directions obtained after demagnetization were randomly scattered, 11 sites had to be rejected. These included green sandstones of the Lettenkeuper, Carnian yellow dolomites, the Rote Wand, the Kiesel- and Stubensandstein and part of the Hettangian–Sinemurian limestones.

According to thermal demagnetization of NRM and of three-component IRM (Lowrie 1990), several categories of magnetic carriers can be distinguished in the remaining sites.

(1) *High unblocking temperature, high coercivity* magnetizations occur in red or yellow to red sandstones (NST, CHS1), in reddish dolomites (CHM) and in grey to red marls interbedded with gypsum (ENT1, ENT2, OSH1, OSH2) (Figs 2 and 3; Table 1). The maximum unblocking temperature of 680 °C indicates that the mineral that accounts for the whole IRM, except that of sites NST and ENT2, is fine-grained haematite. This was expected in the field due to the reddish colour of the rocks.

(2) *Intermediate unblocking temperature, low coercivity* components were observed in the Induan sandstones (NST), in part of the Rhaetian sandstones specimens (SNP, STG, PFD1) and in several beds of the Xeuilley quarry (XEU1, XEU2; Figs 2 and 3). The maximum unblocking temperature of 580 °C and a rapid saturation on IRM curves clearly indicate magnetite as the carrier of the remanence.

(3) *Low unblocking temperature, low to intermediate coercivity* magnetizations were found in most of the rocks that were not completely oxidized [see point (1)]. As will be shown later, most of these magnetizations are believed to be overprints. Maximum unblocking temperatures in the range 300–400 °C suggest the presence of magnetic iron sulphides, particularly in black limestones. In limestones the magnetizations and the susceptibility mostly increase above 350 °C and the directions become scattered, indicating oxidation of the sulphides during heating.

(4) *Low unblocking temperature, high coercivity* components,

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Table 2. Palaeomagnetic results from Triassic–Lower Jurassic rocks of NE France and SW Germany. Bt: maximum unblocking temperatures; N: number of specimens; D/I: declination/ inclination *in situ*; Dc/Ic: declination/inclination corrected for dip; k/α95: Fisher statistic parameters; °N/°E: latitude /longitude of the VGP.

Site	Locality	F,G	Rock type	Statigraphy	Bt	N	D	I	Dc	Ic	k	a95	°N	°E
NSTa	Niedersteinbach	F	red sandstones	Indusian	570-680	16	27	38	27	38	102	4	55	140
NSTb	" " " " " " " "	F		Indusian	540-580	12	16	54	16	54	73	5	71	142
ESA	Eschenau	G	dolomites	Ladinian	350	4	23	63	23	63	716	3	73	111
OSH1	Obersontheim	G	red/green marls	Carnian	680	14	210	-32	210	-32	222	3	51	141
OSH2	" " " " " " " "		red marls	Carnian	680	3	37	34.5	37	34.5	116	11	48	132
ENT1	Entringen	G	grey/red marls	Carnian	680	11	36	40	36	40	213	3	52	129
ENT2a	" " " " " "		red marls	Carnian	680	7	223	-38	223	-38	216	4	46	122
ENT2b	" " " " " "		red marls	Carnian	350	5	11	52	11	52	514	3	72	158
ETD	Eutendorf	G	sandstones	Carnian	400	7	24	65	24	65	104	7	74	98
CHS1	Château-Salins	F	yellow/red sandst.	Carnian	680	8	210	-22	210	-22	25	11		
CHS1a	" " " " " "		" " " " " "	Carnian	680	3	223	-36	223	-36	91	13	45	121
CHM	Charmes	F	red dolomites	Carnian	680	13	42	50	36	52	626	2	54	110
BEG	Berglen	G	red nodules	Norian	680	1	217	-36	217	-36				
SNPa	St. Nicolas du P.	F	sandstones	Rhaetian	>350	16	225	-34	228	-38	convergence		43	114
SNPb	" " " " " " " " " "		sandstones	Rhaetian	100-350	13	50	39	54	43	51	6	42	106
STG	Steingart	G	sandstones	Rhaetian	300-400	15	34	44	35	49	52	5	58	121
PFD1a	Pfrondorf	G	sandstones	Rhaetian	>450	21	219	-54	219	-54	convergence		58	110
PFD1b	" " " " " " "		sandstones	Rhaetian	300-450	19	47	49	47	49	59	4	50	111
PFD2	Pfrondorf	G	black limestones	Hettangian	>400	9	16	64	16	64	264	3	79	112
XEU1	Xeuilley	F	black limestones	Hettangian	580	6	39	58	39	58	150	5	60	101
XEU2	" " " " "		black limestones	Hettangian	580	16	221	-57	221	-57	58	5	59	101
XEUb	" " " " "		black limestones	Hettangian	250-350	30	25	57	25	57	55	4	69	117
CHS2	Château-Salins	F	black limestones	Hettang/Sinemur.	300-400	14	60	53	54	52	54	5	47	98

KGM	Mean Carnian (OSH1+OSH2+ENT1+ENT2a)	4			193	7	49	131
RSD1	Mean Rhaetian (SNPa+SNPb+STG+PFD1a+PFD1b)	5			91	8	50	112
RSD2	" " " " " " " " " (SNPa+SNPb+STG+PFD1a+PFD1b+CHM)	6			109	6	51	112
LEF	Mean Hettangian-Sinemurian (XEU1+XEU2+CHS2)	3			123	11	55	100
EJO	Mean E. Jurassic overprints (NSTb+ESA+ENT2b+ETD+PFD2+XEUb)	6			125	6	74	124

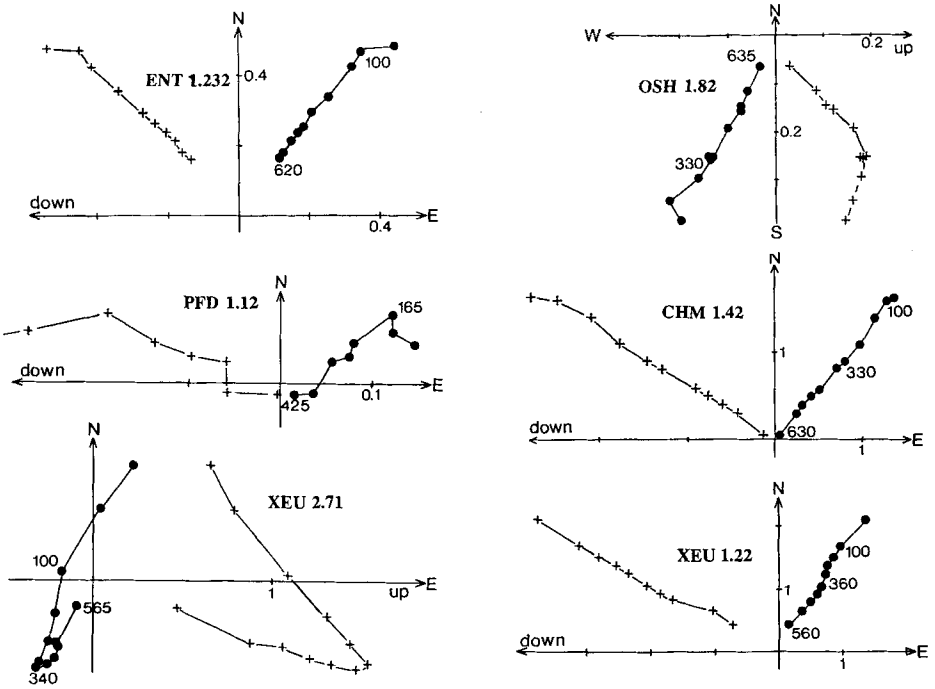


Figure 2. Typical thermal demagnetization of Keuper marls (ENT 1.232 and OSH 1.82), Rhaetian sandstones (PFD 1.12), red dolomites (CHM 1.42) and Hettangian limestones (XEU 2.71 and XEU 1.22). Orthogonal projections in the horizontal (dots) and vertical (crosses) planes. Intensities are in mA m^{-1}.

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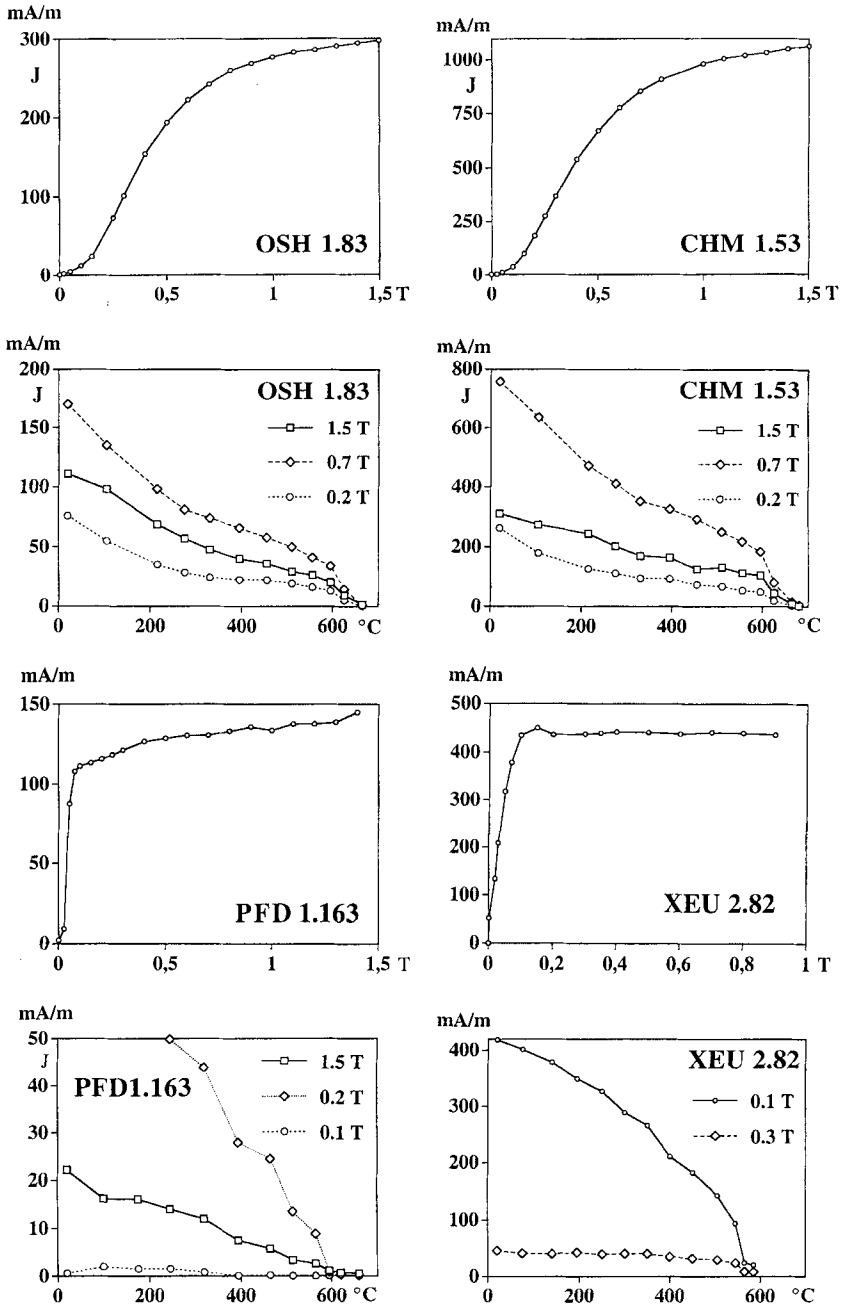


Figure 3. IRM curves and thermal demagnetization of two- or three-component IRM given at high, medium and low fields of marl (OSH 1.83), red dolomite (CHM 1.53) and black limestone (XEU 2.82) samples.

which are removed in the range 50–150 °C, reveal goethite as the magnetic carrier. They occur in part of the Rhaetian sandstones characterized by a yellow–orange colour and in some specimens of black limestones. They often coexist with the previous *low unblocking temperature, low–intermediate coercivity* components, and the directions are sometimes the same.

Palaeomagnetic directions

The demagnetization process has shown univectorial components in most of the red rocks containing fine-grained haematite as magnetic carrier (ENT 1.232, CHM 1.42; Fig. 2). In sites NST, ENT2, XEU1 and XEU2 the Zijderveld plots exhibit two characteristic magnetizations with different unblocking temperatures (XEU 2.71; Fig. 2). The respective directions show two separate clusters of directions (NST a and b, ENT2 a and b; Fig. 4).

The only data that need some filtering and further analysis of the directions are those from the Rhaetian sandstones and from the Schilfsandstein. In the Rhaetian sandstones, some of the main components are well clustered in the northeastern quadrant, while others are very scattered (STG; Fig. 4, crosses). The latter are considered as composite components, which cannot be separated, resulting from the addition of normal and reversed magnetizations with close unblocking temperatures. These intermediate directions were rejected and a mean

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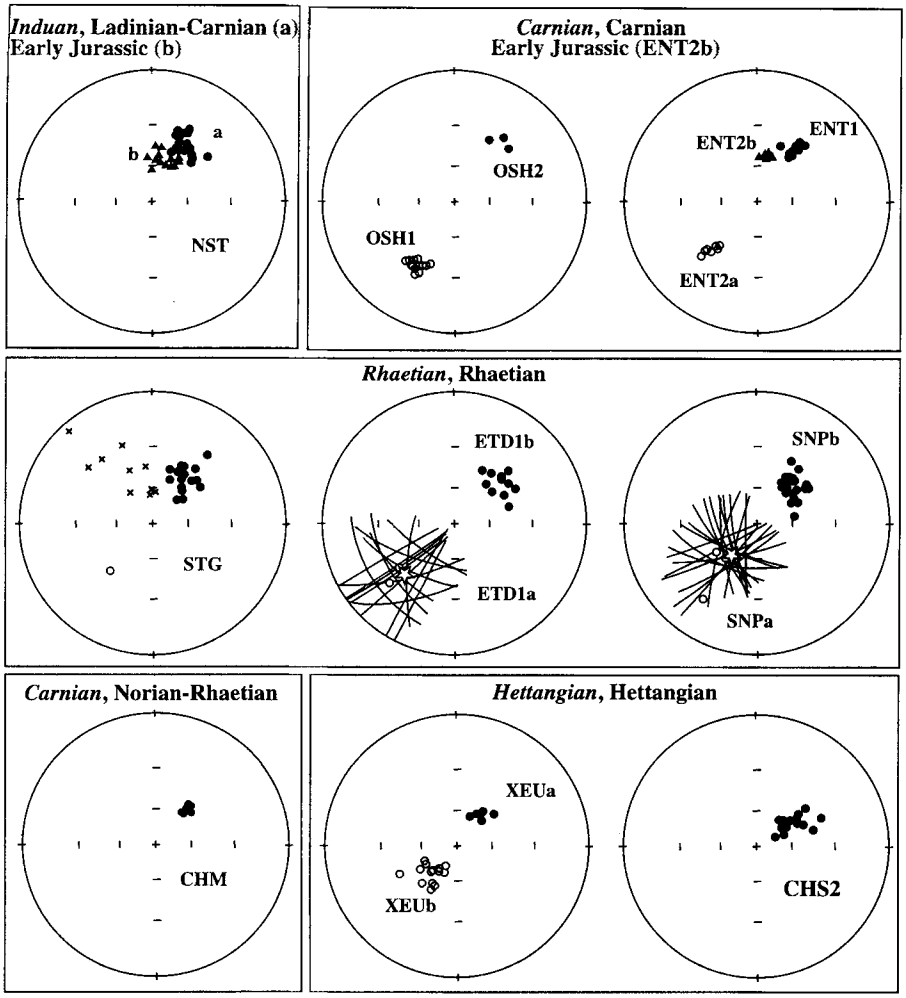


Figure 4. Stereographic projections of the characteristic directions yielded after thermal demagnetization. Full circles are in the lower hemisphere, open circles in the upper hemisphere. Triangles indicate overprints. Converging circles and the mean directions (small stars) calculated from converging circles are plotted for sites ETD1 and SNP. Crosses (STG) represent composite directions discarded from the mean calculation.

direction was calculated with the clustered northeasterly directions. In most cases the main component does not pass through the origin, which suggests the presence of a residual magnetization with higher unblocking temperature (PFD 1.12; Fig. 2). In two of the three Rhaetian sites, it was possible to apply the converging circles method (ETD and SNP; Fig. 4), and a direction with reversed polarity was calculated. In a few samples the existence of this reversed magnetization is confirmed directly on the Zijderveld plot. Concerning the Schilfsandstein of site CHS1, a hard secondary magnetization aligned with the present-day direction could not be removed completely by heating. Most remaining directions are defined by endpoints, and only a few by vectors. As the primary magnetization has a reversed polarity, the ill-defined directions show a large scatter in inclination. Some directions are consistent with the expected Carnian direction but most are too shallow. Therefore, the mean direction CHS1 was not kept for the calculation of mean directions.

When considering the most reliable results, the mean directions obtained in the Triassic–Early Jurassic units look scattered (Fig. 5). The declinations range from 11 to 54°, while the inclinations vary from 22 to 64°. The polarity is exclusively normal for the directions with low declinations (11 to 25°) and high inclinations (52 to 64°), and is both normal and reversed for the directions with more easterly declinations and lower inclinations. The stratigraphic sequences are nearly flat lying, precluding any fold test. The criterion of reliability will be the occurrence of reversals in the sedimentary sections. As far as the directions with only normal polarity are concerned, an attempt will be made to date them by correlating the poles with available results from Europe. The results will be presented and discussed according to the stratigraphic positions of the sites.

Induan (Annweiler) sandstones

According to its stratigraphic position (Lower Buntsandstein = Induan), the VGP of site NST is expected to be located between the Thuringian poles MAS and SAT of southern France (Merabet & Daly 1986; Diego-Orozco & Henry 1992) and the Lower Anisian (upper Buntsandstein) pole of the northern Vosges (Fig. 6a) (Edel 1993). Two components (NSTb and NSTa, Fig. 4), removed at 550–580 and 680 °C, respectively, were isolated during heating. The high-temperature component NSTa, which is carried by haematite, has a pole position close to the Ladinian (Muschelkalk) pole of Heming

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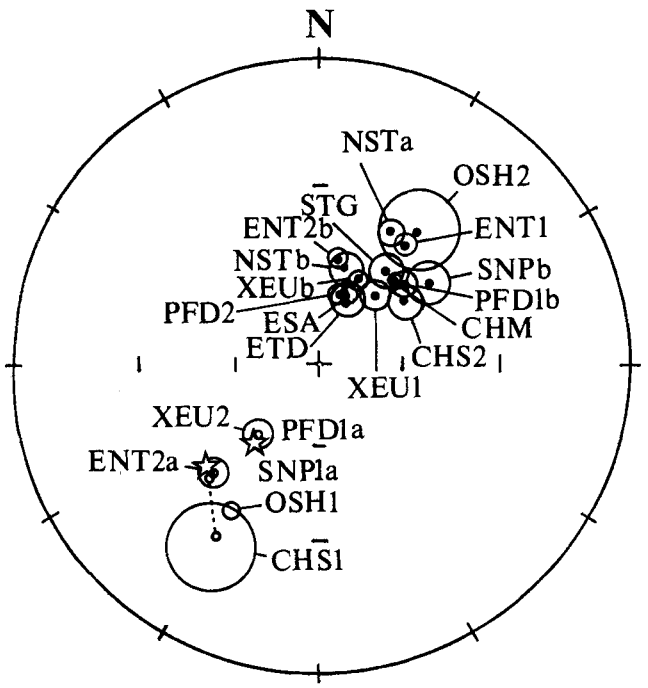


Figure 5. Stereographic projection of the mean site directions with α_{95} confidence circles. Stars represent directions calculated from great circles.

(HEM) (Théveniaut *et al.* 1992) and the new results obtained from the Keuper marls (OSH and ENT, see below). The intermediate-temperature component NSTb has a high inclination and a more northerly declination, which locates the VGP close to the mean Lower Jurassic pole of Europe (Van der Voo 1990). This suggests that the Lower Triassic sandstones have probably undergone two overprinting phases, the first in mid-Triassic times and the second in the Lower Jurassic. A mean direction also consistent with the Lower Jurassic results of Europe was obtained using the low-unblocking-temperature magnetizations in yellow dolomites from the Lettenkeuper (Eschenau quarry, ESA; Table 2).

Carnian marls (Gipskeuper)

Sampling was undertaken in gypsum quarries in two localities, Obersondheim (OSH) and Entringen (ENT), which are 85 km apart (Fig. 1). Red and occasionally grey-green marls are interbedded with the gypsum layers. The magnetization is restricted to the red marls, while the white lenses of gypsum proved not to be magnetic. Thermal demagnetization of marls yielded very stable magnetizations and narrow clusters of reversed and normal directions (OSH1, OSH2, ENT1, ENT2a, Figs 2, 4 and 5). IRM curves and thermal treatment of the NRM, of the IRM and of the total magnetization suggest that the carrier of the remanence is fine-grained haematite (ENT 1.232, OSH 1.82, Fig. 2; OSH 1.83, Fig. 3). In the Obersondheim quarry, the normal-polarity OSH1 was measured in the mostly grey-green marls of the upper beds [facies cycle 11 of the Aigner & Bachmann (1989) sequence], while the reversed-polarity OSH2 occurred in the red marls of the lower beds (cycle 5). With respect to α_{95}, the mean directions are antiparallel ($\gamma = 6°$, angle between OSH1 and OSH2 plotted in the same hemisphere; Figs 4 and 5). At the Entringen locality a bed of red marls (30 cm thick) in the northerly quarry (ENT2), and two beds of grey-red marls (60 cm thick) were sampled in the uppermost part of the southerly quarry (ENT1). After heating, specimens of site ENT1 exhibited well-clustered directions with normal polarity. On the contrary, the reddish specimens from site ENT2 showed a cluster of directions with reversed polarity (ENT2), which were occasionally superimposed by a low-temperature component of normal polarity and higher inclination (ENT2b). Again, the mean directions ENT1 and ENT2a are nearly antiparallel ($\gamma = 6°$; Figs 4 and 5).

The marl and gypsum beds of the Entringen quarries are flat lying. Those of the Obersondheim quarry show slight undulations, with an amplitude of about half a metre, due to differential compaction. These folds having no repercussions in the directions OSH1 and OSH2 (negative fold test); the magnetizations are considered not as detrital but as secondary in origin. Under the optical microscope, the marls and dolomites are masked by the omnipresence of gypsum. It occurs in three forms, as follows.

(1) Millimetric layers of fine-grained sugary gypsum in a stratiform position devoid of clay. Its primary origin is possible but unlikely.

(2) Fibrous gypsum with fibres perpendicular to the bedding plane and totally impregnated through the marls. The ratio of gypsum to marls changes rapidly laterally and vertically. Its diffuse impregnation in the marls and its 'non-stratiform occurrence' support a secondary crystallization.

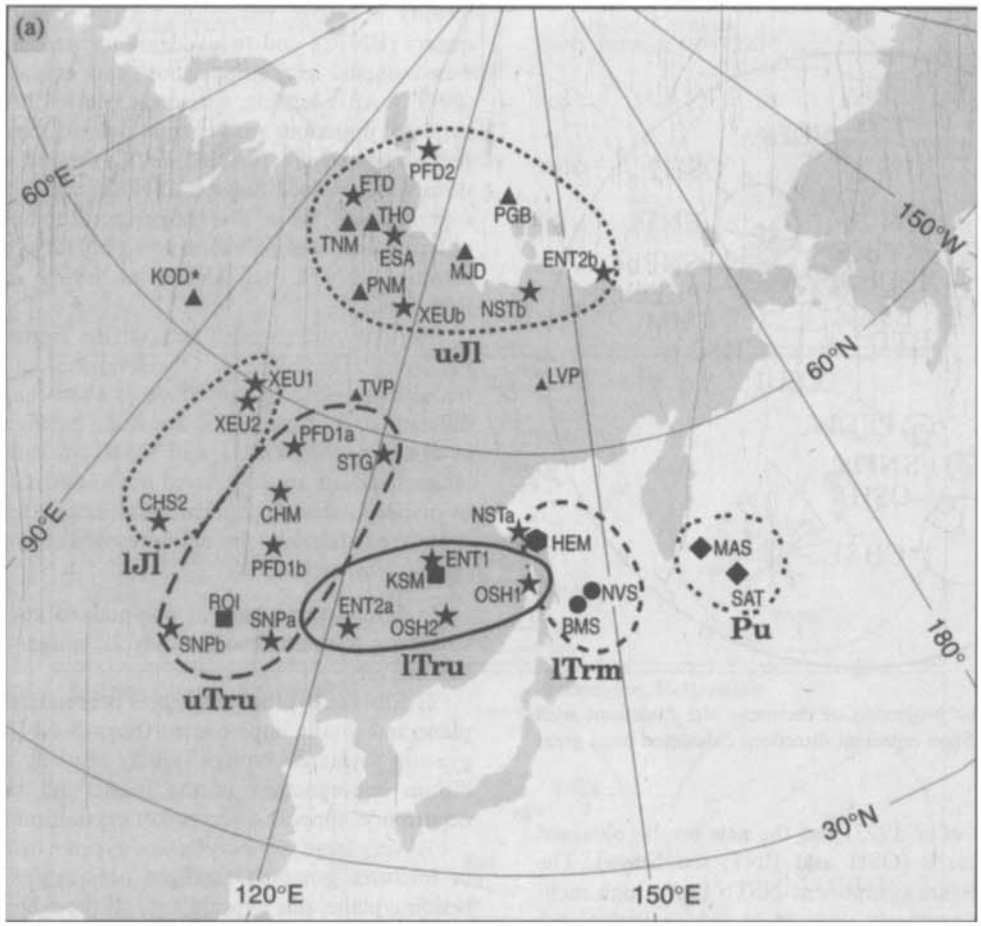
(3) Very large monocrystalline gypsum inside fissures, cracks or fractures generally arranged obliquely with regard to the bedding plane; this gypsum cuts all the other structures and is typically the last formation of gypsum. Water escape structures occur locally with diagenetic fractures in the marls which have a Christmas tree pattern. The macroporosity is filled with structureless clay, which shows secondary formation of gypsum. These synsedimentary or early diagenetic structures are related to water or brine escape during sediment compaction.

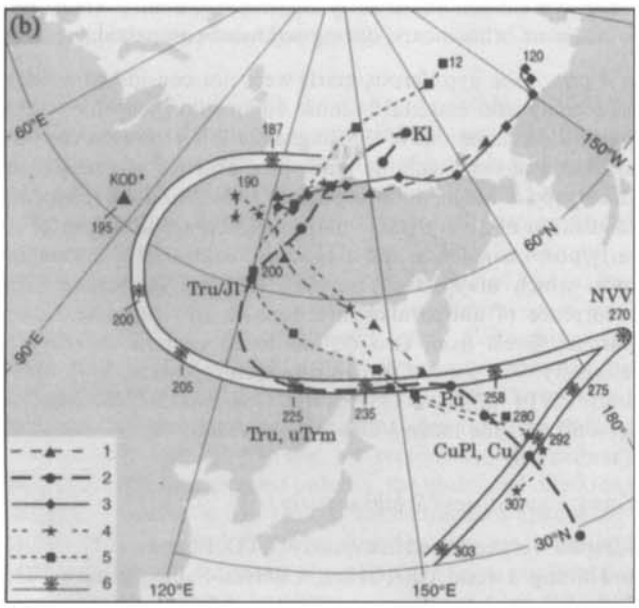
A priori, the gypsiferous marls were not considered as ideal palaeomagnetic material because compaction may have shallowed the inclinations of the magnetization. However, the very small size of the haematite grains, the absence of a significant anisotropy of magnetic susceptibility (AMS) and the secondary acquisition of a very stable magnetization are in favour of an early/post-compaction and diagenetic acquisition of the remanence, which may have preserved the real inclination. The occurrence of antiparallel directions in two separate Lower Carnian levels from two distant localities is a criterion of reliability. The mean Carnian direction calculated with directions ENT1, ENT2a, OSH1 and OSH2 is 36°/36° ($k = 209$, $\alpha_{95} = 6°$) and the mean VGP is 49°N/131°E.

Carnian sandstones (Schilfsandstein)

Samples were collected in a quarry (ETD, Eutendorf, Germany) and along a road cut (CHS1, Château-Salins, France). The dark, brownish and coarse-grained sandstones of Eutendorf only exhibited a low-temperature component with a mean direction 24°/65°, similar to the Lower Jurassic results from Europe (Van der Voo 1990; Beck 1994). Concerning the yellow to red sandstones CHS1, a hard secondary magnetization aligned with the present-day direction could not be eliminated

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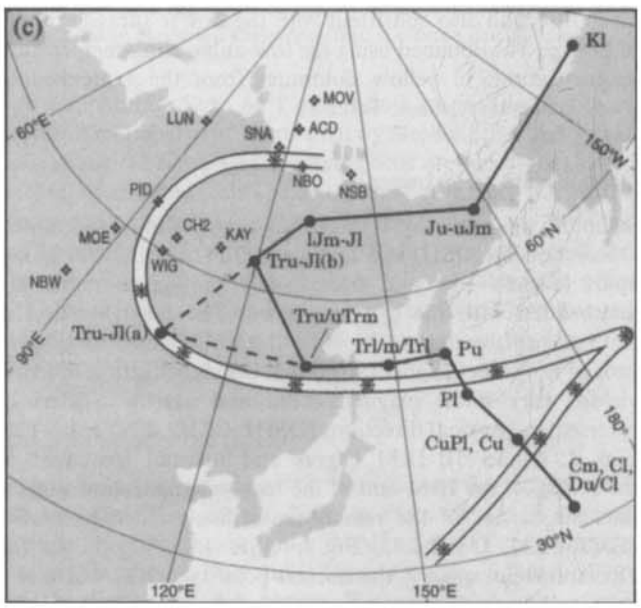


Figure 6. (a) Palaeomagnetic poles from this study (stars) and published poles for the Upper Permian–Lower Jurassic of southern 'stable' Europe (Table 3). (b) Published APWPs after: (1) Westphal *et al.* (1986), (2) Van der Voo (1990), (3) Besse (1986), (4) Théveniaut (1993) and (5) Beck (1994). (6) Revised APWP according to Triassic–Lower Jurassic poles from Table 4. (c) APWP of North America (Van der Voo 1990) and selected poles that deviate from the APWP.

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completely. As a consequence, the inclinations scatter from $-4°$ up to $-35°$, and the inclination of the mean direction ($210°/-22°$) is too low with respect to the expected inclinations, which should be higher than those of the Anisian sandstones (Edel 1993) and the Ladinian limestones (Théveniaut *et al.* 1992). The three specimens that show the highest inclinations have a mean direction $223°/-36°$, which is nearly the same as the direction ENT2a obtained in the Lower Carnian marls. Nevertheless, the direction CHS1a is poorly documented and will not be considered in the discussion. The results look similar to those obtained by Hahn (1984) on reed sandstones of Carnian age from Swabian, i.e. (1) reversed polarity, (2) incomplete removal of hard secondary components with normal polarity, and consequently (3) ill-defined characteristic magnetization.

Carnian dolomites

The generally ochre to beige dolomites overlie the reed sandstones at Charmes (CHM). Sampling was performed at different places but was only successful in one level of dolomites characterized by a reddish pigmentation. The magnetizations are remarkably stable and the directions cluster closely around the mean $42°/50°$ ($k=626$, $\alpha_{95}=2°$) (Figs 2, 4 and 5). A direction with reversed polarity is exhibited by one sample with an ochre colour. The intensity of the corresponding magnetization is up to 100 times lower than that of the red samples, which is around 5 mA m^{-1}. IRM procedure and thermal treatment of red samples show the same characteristics as for the red marls and reveal fine-grained haematite as the only magnetic carrier (CHM1.53, Fig. 3).

A unique but well-defined magnetization, carried by haematite and with a direction $217°/-36°$, consistent with the results from Carnian marls (ENT1–2a and OSH1–2) and sandstones (CHS1b), was obtained in a red nodule within the Stubensandstein (Norian sandstones: BEG, Table 2).

The stratigraphic position of the red dolomites leads to the expectation of a palaeomagnetic direction intermediate between OSH–ENT (Carnian) and the unique direction from BEG (Norian). This is not the case, and the mean *in situ* direction lies in the vicinity of the directions measured in the Rhaetian sandstones (see below). At site CHM the rock consists of pale yellow dolomite in the form of decimetric strata. Some dolomitic strata have a very discontinuous pale red or rose colour that cross-cuts the bedding plane. Well-developed stylolites are associated with this local colouration. The stylolites are, moreover, perpendicular to the bedding plane (compaction–dissolution). Analysis of thin sections reveals the following chronology:

(1) deposition of carbonates;
(2) first dolomitization phase leading to macrorhombohedrons of dolosparite;
(3) second dolomitization phase characterized by dolomicrosparite;
(4) centrifugal diffusion of very small red grains of iron oxides from well-individualized opaque grains.

As the magnetic properties indicate that the magnetization is carried by small grains of haematite, the best candidates are these secondary red oxides. The magnetization in the red dolomites being secondary, the age of the overprints and the dip of the series during overprinting are unclear. During the Jurassic, the environment of the area investigated was marine. Consequently, the oxidation process leading to the red pigmentation must have occurred before the Jurassic, in the Upper Norian or Rhaetian. Preference is given to a post-tilting acquisition of the magnetization because of the consistency of the direction with the mean direction obtained in Rhaetian sandstones.

Rhaetian sandstones

Two quarries were sampled in the vicinity of Tübingen (STG and PFD, Germany) and a third to the east of Nancy (SNP, France). Thermal treatment showed a multicomponent behaviour of the magnetizations. A significant drop of up to 40 per cent of the NRM occurred in a large part of the specimens at 100–150 °C. The main magnetization is characterized by low unblocking temperatures in the range 300 to 450 °C and by normal polarity (PFD1.12, Fig. 2). On Zijderveld plots most of these vectors do not join the origin and reveal the existence of a harder magnetization with reversed polarity. This is confirmed in each of the three sites by a few specimens that show a component with intermediate unblocking temperatures of 500–590 °C with reversed polarity (Fig. 4). Demagnetization of three-component IRM in specimens from PFD1 shows a decrease of the high-coercivity component (1.5 T) at 100 and 580 °C (PFD1.163; Fig. 3). The high coercivity of the component removed at 100 °C suggests goethite as a carrier. Goethite may be responsible for the orange colouration of part of the sandstones. The direction of this component often has the same direction as the main component, with unblocking temperatures in the range 300–450 °C. The intermediate-coercivity (0.2 T) component, which accounts for the major part of the IRM, is demagnetized at 580 °C, and indicates magnetite (Fig. 3). A decrease in the range 350–400 °C appears on the intermediate-coercivity component, while the weak, low-coercivity (0.1 T) component vanishes at these temperatures. It is unclear whether the low- or the intermediate-temperature magnetization is an overprint. Two explanations may be given: (1) primary sulphides were partly oxidized to magnetite; (2) primary magnetite was partly altered, producing maghemite. In a few specimens a small residual remains above 580 °C, suggesting the presence of haematite. In two sites (PFD1 and SNP) the converging circles method could be applied (Halls 1976). The reversed mean directions obtained by this method are consistent with the few directions isolated on Zijderveld diagrams. Magnetite is the carrier of these reversed magnetizations. The secondary magnetizations were not acquired a long time after the primary ones because the respective directions are nearly antiparallel ($\gamma=6°$ between SNPAa and SNPAb; $\gamma=7°$ between PFD1a and PFD1b), and the mean VGP of the components with reversed polarity (50.5°N/112°E, $K=57$, $N=2$) is the same as the mean VGP of the components with normal polarity (50°N/112°E, $K=76$, $N=3$).

Hettangian–Sinemurian limestones

Hettangian black limestones were sampled (1) on top of the sandstones in the Pfrondorf quarry (near Tübingen) (PFD2), (2) in the lower layers of the limestone and marl quarry of Xeuilley (XEU), south of Nancy, and (3) along a road cut near Château-Salins (CHS2) (Fig. 1). At sites XEU and CHS2, the black limestones and marls contain abundant *gryphaea.*

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Thermal demagnetization reveals characteristic directions with low unblocking temperatures of up to 350 °C in all three localities, and all magnetizations show a normal polarity. In the Xeuilley quarry, the following sequence occurs above the Rhaetian–Hettangian boundary:

(1) the first metre of limestones contains low unblocking temperature magnetizations with normal polarity;

(2) in the next 60 cm of limestones, intermediate unblocking temperature magnetizations with normal polarity (XEU1) occur in addition to low unblocking temperature magnetizations of normal polarity (XEU 1.22, Fig. 2);

(3) in the next metre, demagnetization exhibited intermediate unblocking temperature magnetizations with reversed polarity in addition to low unblocking temperature magnetizations with normal polarity (XEU2.71, Fig. 2);

(4) in the following beds, up to 8 m from the bottom, only low unblocking temperature magnetizations with normal polarity were isolated.

Rapid saturation (0.1 T) and complete removal of the magnetization at 580 °C identified the carrier of the intermediate-temperature magnetization as magnetite (XEU2.82, Fig. 3). Low unblocking temperatures and low- to intermediate-coercivity minerals are interpreted to be iron sulphides. This was expected in the field due to the distinctive smell of the rock during drilling or breaking. In these rocks the susceptibility increases strongly above 350 °C, indicating oxidation of the sulphides, probably to magnetite.

The mean directions XEU1 and XEU2 that are carried by magnetite are very stable and fairly antiparallel ($\gamma = 1°$; Figs 4 and 5). The corresponding VGPs occupy a position intermediate to the mean VGP from Rhaetian sandstones and the result from the Kerforne Dyke (Sichler & Perrin 1993). The poles of the low unblocking temperature components (CHS2, XEUb and PFD2) show an elongated distribution from a position slightly east from the Rhaetian pole and the Lower Jurassic poles of Europe (Fig. 6a). Within the Xeuilley quarry, the directions of the low unblocking temperature components are scattered from directions consistent with the directions carried by magnetite (XEU1) to northerly directions with higher inclinations, so that the mean VGP XEUb has a position intermediate to XEU1 and PFD2 (Fig. 6a). Their location within the uJl group means that the low unblocking temperature components are mainly overprints.

DISCUSSION

The Carnian marls are characterized by antiparallel, reversed and normal directions of diagenetic origin. The mean pole KGM lies very close to the KSMn pole corresponding to the directions with normal polarity obtained by Creer (1959) from Keuper marls from Sidmouth (SW England). The directions with reversed polarity measured in these marls are characterized by significantly lower inclinations and a greater dispersion than those of normal polarity, and, consequently, a negative reversal test. The reason for these low inclinations and greater dispersion is probably the same as that for the sandstones of Château-Salins (CHS1), i.e. incomplete elimination of the secondary magnetization with normal polarity. Contamination by a component with normal polarity and high inclination affects the inclination of a magnetization with reversed polarity significantly more than a magnetization with normal polarity. Giving the same weight to the KSM pole as to an individual result from the German Keuper marls, a mean pole (50°N/131°E) was calculated for the lower Upper Trias (Table 4).

Primary and secondary magnetizations with nearly antiparallel directions were observed in the Rhaetian sandstones. After correction for a slight tilt, the scatter of the poles does not change ($K_c = K_0 = 91$) and the fold test is inconclusive. The latitude of the mean pole remains the same (50°) and declination turns from 116 to 112°E. The pole CHM obtained from *in situ* directions in red dolomites is very close to this mean pole. The oxidation process leading to secondary haematite in the dolomites probably occurred in the Norian–Rhaetian, before the Lower Jurassic marine transgression. A mean pole (51°N/112°E) including the results from the sandstones and the CHM pole was computed for the Latest Trias. Concerning previous investigations, the pole ROI from the Rochechouart impact (Pohl & Soffel 1971) falls into the group of poles obtained in the Rhaetian sandstones (Fig. 6a). Scattered ages of 210 ± 100 Ma (fission track; Storzer, in Pohl & Soffel 1971), 186 ± 8 Ma (Grieve 1991) and 163 ± 10 (K–Ar; Grieve *et al.* 1985), preclude the inclusion of the result in the calculation of the mean Latest Triassic pole. Thus, the consistency of the ROI pole with the mean Rhaetian pole would favour an Upper Triassic age around 210 Ma.

The stable antiparallel directions, carried by magnetite, which occur in different beds of the Xeuilley quarry, provide a reliable pole (60°N, 101°E) for the Lower Lias. When including the pole corresponding to low-temperature magnetizations in CHS2, the mean pole becomes (55°N/100°E). The new pole is not far from the result from the Kerforne doleritic dyke (Brittany) recalculated after elimination of the deviated direction B4 (Table 3) (Sichler & Perrin 1993). As a consequence of this correction, the k factor increases from 36 for KOD ($N = 7$) to 69 for KOD* ($N = 6$) (Table 3).

Northeast of the Hettangian pole and north of the Rhaetian pole is the pole TVP obtained from Upper Triassic–Lower Jurassic ophitic igneous rocks from the northern Pyrenees (Girdler 1968). This result, together with the result from Hettangian tuffs (LVP), comes from a strongly tectonized area with steep, even overturned, bedding. Possible errors in declination due to tilted fold axes cannot be excluded; therefore, the results have not been included in the calculation of the mean poles.

Six poles computed from mostly low unblocking temperature components present in various stratigraphic units (NSTb: Induan; ESA, ENT2b and ETD: Carnian; PFD2 and XEUb: Hettangian) fall into the main group of Liassic poles (Jl, Fig. 6a). The mean pole EJO computed from all six mean directions (74°N/124°E) is consistent with the mean pole (73°N/116°E) calculated from Lower Jurassic results from Europe obtained from Pliensbachian and Toarcian sediments from the UK (PGB: Hijab & Tarling 1982), France (PNM, TNM: Fabre 1986; THO: Galbrun, Gabilly & Rasplus 1988) and the Messejana Dyke in Iberia (Schott, Montigny & Thuizat 1981) (Table 4). The later pole was recalculated after closing the Bay of Biscay using a rotation pole located in the corner of the Gulf of Gascogne and an angle of rotation of 35° (Edel 1980; Sibuet & Colette 1991). The consistency of both mean poles argues for an overprinting phase of part of the Triassic rocks during the Lias. Most of the overprints show low unblocking temperatures.

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Table 3. List of palaeomagnetic data available for western Europe for the Upper Permian–Lower Jurassic time-span. Pol: polarity of the magnetization. *: VGP recalculated using a pole of rotation for Iberia at 44°N/2°W and an angle of 35° (Edel 1980; Sibuet & Colette 1991).

Symb.	Rock unit Locality	Stratigr. level	M. age Ma	Lat °N	Long °E	K, k	a95	Pol.	Reference
THO	limestones, Thouars, F.	Toarcian	183	73	105	30	4		Galbrun et al., 1988
TNM	limestones, Normandie, F	Toarcian	183	72	100	257	8	N+R	Fabre, 1986
PNM	limestones, Normandie, F	Pliensbachian	190	68.5	118	94	11	N	Fabre, 1986
PGB	sediments, U.K.	Pliensbachian	190	77	135	37	2		Hijab and Tarling, 1982
MJD	Messejana dyke, SP	Hettang.-Sinemur.	195	73*	126*	36	7	N	Schott et al., 1981
LVP	volcanics, Pyrenees, F	Hettangian		65	144	70	7	N	Girdler, 1968
KOD	Kerforne dykes, Brittany, F	Sinemurian	195	61	79	36	10	N	Sichler and Perrin, 1993
KOD*	" " " " " " " "	" " " " " " " "		62	86	69	8		idem minus 1 direction
ROI	Rochechouart Impact, F.		210?	44	110	310	4	R	Pohl and Soffel, 1971
TVP	volcanics, Pyrenees, F	Keuper		62	114	70	7	N	Girdler, 1968
KSMn	marls, Sidmouth, U.K.	Keuper	225	51	130			N	Creer, 1959
KSMr	" " " " " " "	" " " " "		37	126			R	" " " " "
BMS	sediments, E. Germany	Bunts./Muschelk.	230	49	146	15	15		Rother, 1971
HEM	limestones, Heming, East F	Ladinian	235	53.5	141	97	3	N+R	Théveniaut et al., 1992
NVS	sandstones, N. Vosges, East F	Indusian	240	50	148	614	3	R+N	Edel, 1993

Table 4. Mean poles for western Europe, for the Upper Carboniferous–Lower Jurassic time-span.

Period	Age, Ma	Lat. °N	Long. °E	n/N	k/K	α/A95	References or poles from tab.2 & 3
uJl	187	73	117	6	304	4	MJD+PGB+PNM+TNM+THO+EJO
lJl(2)	195	62	86	6	69	8	KOD*
lJl(1)	200	55	100	3	123	11	XEU1+XEU2+CHS2
uTru	205	50	111	5	116	7	ROI+SNP+STG+PFD1+CHM
lTru	225	50	131	5	252	5	KSMn+OSH1+OSH2+ENT1+ENT2a
lTrm	235	51	145	3	619	5	BMS+HEM+NVS
Pu,uPm	258	49	167	13	102	4	Edel and Schneider, 1995
mPl	270	45	187	3	675	5	" " " " " " " " " " " " " " " " "
mPl	275	44	177	7	81	7	" " " " " " " " " " " " " " " " "
lPl	292	41	169	21	170	2	" " " " " " " " " " " " " " " " "
uCu	303	32	153	8	664	2	" " " " " " " " " " " " " " " " "
lCu	320	8	123	17	18	8	Edel and Lewandowski, 1993

The Upper Carboniferous to Lower Jurassic APWP of Europe

A recent study of Upper Carboniferous–Lower Permian volcanic units from the northern Vosges and the Schwarzwald has led us to revise the Upper Carboniferous to Upper Permian APWP of the European plate (Edel & Schneider 1995). Overprints recorded during the major part of the Permian show a cusp in the APWP in Upper Autunian–Lower Saxonian times (Fig. 6b). Due to the calculation of mean poles from results that probably include overprints, acquired over a long period of time, the previous APWPs smoothed the sharpness of the cusp (Van der Voo 1990; Théveniaut 1993; Beck 1994) (Fig. 6b).

The new results suggest a similar feature for the Upper Triassic–Lower Jurassic part of the APWP. With the exception of the mean Carnian pole, which is still on Van der Voo's (1990) APWP, the mean poles computed for the Rhaetian and the Hettangian are at odds with previously published APWPs. They lie in the continuation of a westward track from the mid-Permian to mid-Triassic part of the APWP at a latitude which remains around 50°N for about 70 Myr. The new Upper Triassic–Lower Jurassic poles shift the position of the loop on the published APWP by about 25° towards the west. The loop began at the end of the Rhaetian and finished in the Pliensbachian. The corresponding global tectonic event lasted for the duration of the Lower Lias, i.e. for 10–15 Myr.

The North American APWP in the light of the revised APWP of Europe

Using the Bullard, Everett & Smith (1965) reconstruction, the rotated North American APWP (Van der Voo 1990) and the revised European APWP fit together from the Upper Permian to the Upper Triassic–upper Middle Triassic, and up to the Upper Triassic–Lower Jurassic when taking the high-quality data from Van der Voo (1990; $Q>4$) (Fig. 6c). Previously, Gordon, Cox & O'Hare (1984) proposed a two small-circles track with a cusp at about (61°N/96°E) for the Upper Carboniferous–Cretaceous APWP of North America. This

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cusp is supported by various results from the Chinle and Wingate formation (Reeve 1975; CH2, WIG, Fig. 6c), the Kayenta formation (Steiner & Helsey 1974; KAY), Piedmont Dykes (Smith 1987; PID), the Moenave formation (Ekstrand & Butler 1989; MOE) and the Newark Basin, West Limb (Van Fossen, Flynn & Forsythe 1986; NBW). This group of poles is in agreement with the Hettangian pole from this study as well as with the revised pole from the Kerforne Dyke (Sichler & Perrin 1993). Because of its high latitude, the recalculated Upper Liassic mean pole of Europe (73°N/117°E) does not agree with the mean Lower–Middle Jurassic pole proposed by Van der Voo (1990). However, it supports the hypothesis proposed by Van Fossen & Kent (1990) that the North American APWP ventured to high latitudes in the Middle Jurassic. The group of American poles from dykes of the Anticosti Island (Larochelle 1971; DAI), the Luning and Sil Nakya formations (Kluger-Cohen, Anderson & Schmidt 1986; LUN, SNA, Fig. 6c), overprints in Carnian–Norian strata of the Newark basin (Witte & Kent 1990) (NBO), and from the Moat volcanics (Van Fossen & Kent 1990; MOV) can be superposed on the group of overprints (EJO) of this study and the group of Upper Liassic poles of Europe (Fig. 6a).

CONCLUSIONS

The new results make up for the lack of data from the Lower Triassic to the Lower Jurassic in a satisfactory manner and give some support to previous data from the Rochechouart meteorite impact (Pohl & Soffel 1971) and from the Kerforne Dykes in Brittany (Sichler & Perrin 1993), which up to now were considered as exotic relative to the computed APWPs (Fig. 6). The results can be summarized as follows:

(1) a new pole for the Lower Carnian has been obtained from Keuper marls, with normal and reversed polarities and with haematite as the remanence carrier;

(2) a new pole has been obtained for the Latest Trias (Rhaetian) based on directions with normal and reversed polarities in sandstones and red dolomites;

(3) a new pole has been obtained for the Lower Liassic from Hettangian limestones, based on normal and reversed magnetizations carried by magnetite;

(4) an overprinting phase in Lower Jurassic times lead to mostly low-temperature magnetizations in various rock types from the areas investigated.

A revised APWP based on the new results and a critical review of previous data is proposed. This APWP is in agreement with a large part of the North American data when using Bullard's reconstruction.

The Lower Mesozoic plate motion of Europe

The geodynamical consequences of the revised APWP of the European plate are illustrated in terms of variations of the declination and palaeolatitude at a locality close to Strasbourg with coordinates 47.5°N/7.5°E (Fig. 7). With respect to previous APWPs (Van der Voo 1990; Beck 1994), the plate motions are significantly more pronounced, particularly in the mid-Permian and the Upper Triassic–Lower Jurassic. Subsequent to the late Alleghanian tectonic phase (mid-Permian), which marked the end of the clockwise rotation of Europe and the end of the Variscan orogeny, the motion of

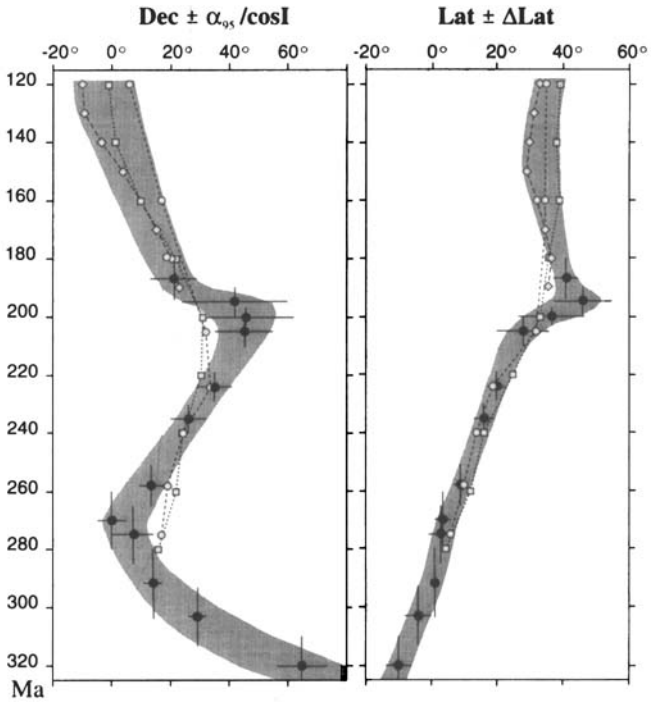


Figure 7. Evolution with time of the declinations and palaeolatitudes for a locality presently at 48.5°N/7.5°E, near Strasbourg, according to this study and Edel & Schneider (1995) (full circles, Table 4), and after the APWPs from Van der Voo (1990) (open dots), Besse (1986) (diamonds) and Beck (1994) (squares).

the European plate became a counterclockwise rotation. The rotation started in the mid-Permian, around 270 Ma (Edel & Schneider 1995), and lasted up to the end of the Trias, around 203 Ma, with a nearly constant rate of 1.4° Myr^{-1} for the rotation of the declination. The northward drift in latitude by 35° remained constant from 320 to 203 Ma. The constant rate of 3.4 cm yr^{-1} during 115 Myr (Fig. 7) increased in the Earliest Jurassic according to the results from the Hettangian limestones and the Kerforne Dyke (Sichler & Perrin 1993). The end of the northward motion of Pangaea around 203 Ma was followed by a rather rapid rotation that corresponds to a change in declination by about 25° at Strasbourg and that lasted about 15 Myr. The cessation of the constant, 70 Myr duration, northward drift of Pangaea and the subsequent Liassic rotation suggest a collision of the megaplate somewhere along its northern border. The Eulerian pole of this counterclockwise rotation is precisely located in northeastern Asia, either in northeastern Siberia or in the Mongol–Okhotsk belt. At the same time, the Atlantic rifting started and numerous doleritic dykes, including the Kerforne Dyke in Brittany (Sichler & Perrin 1993) and the Messejana Dyke in Iberia (Schott *et al.* 1981), intruded both sides of the rift. This synchronism suggests that it was the cessation of the Triassic drift of Pangaea and the rapid change of its geodynamical regime that led to the intraplate fracturing of the megacontinent, and subsequently to the Atlantic rifting.

This study stresses the necessity of improving the accuracy of the APWPs of the continental plates in order to constrain more precisely their geodynamical evolution. Compared to the abundance of palaeomagnetic results from mobile microplates and belts, the data from large plates are too scarce for some

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critical periods. The Jurassic–Cretaceous part of the European APWP in particular requires new data.

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